

ASSIGNMENT 1

HOW THE INTERNET DELIVERS A WEB PAGE

A Technical Study of Accessing Google Search through a Web Browser

Assignment Report

This report explains what happens when a user enters **www.google.com** in a web browser and presses Enter. The process involves several Internet protocols working together across the five-layer TCP/IP model. The report discusses DNS, HTTP, TCP, client-server architecture, cookies, HTTP and FTP, and the complete communication path between the browser and the web server.

Website Chosen: Google Search (www.google.com)

1. TCP/IP PROTOCOL STACK

The TCP/IP protocol stack is a layered model that describes how data is prepared, transmitted, routed, delivered, and received across the Internet. Each layer performs a particular function and works with the layers above and below it.

APPLICATION LAYER HTTP/HTTPS • DNS • FTP • SMTP	Higher-level services
TRANSPORT LAYER TCP • UDP	
NETWORK LAYER IP • ICMP	
DATA LINK LAYER Ethernet • Wi-Fi • PPP	
PHYSICAL LAYER Cables • Fibre • Radio Signals	Physical transmission

Layer	Function	Examples
Application	Provides services directly to applications. HTTP/HTTPS enables web communication, while DNS resolves domain names.	HTTP, HTTPS, DNS, FTP, SMTP
Transport	Provides process-to-process communication. TCP provides reliable delivery, while UDP is designed for low-overhead communication.	TCP, UDP
Network	Provides logical addressing and routing so packets can travel between different networks to reach the destination.	IP, ICMP
Data Link	Transfers frames between directly connected devices and handles MAC addressing, framing, and error detection.	Ethernet, Wi-Fi, PPP
Physical	Carries raw bits through physical media such as copper cables, optical fibre, or wireless signals.	Cables, fibre, radio

2. DNS: CONVERTING THE DOMAIN NAME INTO AN IP ADDRESS

The Domain Name System (DNS) translates a human-readable domain name into an IP address. A domain name such as **www.google.com** is easier for people to remember, while network communication requires an IP address.

DNS Resolution Process:

1. The user enters **www.google.com** in the browser.
2. The browser checks its local DNS cache.
3. If no suitable entry is found, the operating system and DNS resolver are consulted.
4. The DNS resolver obtains the required address from the DNS system.
5. The IP address is returned to the browser.
6. The browser can now contact the Google server using that address.

Domain name → DNS resolver → IP address → Web server

DNS is important because it hides the complexity of numerical IP addresses and allows users to access Internet services using memorable names.

3. HOW HTTP IS USED TO REQUEST AND RECEIVE THE WEBPAGE

After DNS provides the destination address, the browser communicates with the Google web server using HTTP or HTTPS. The browser sends a request for the required resource, commonly using the GET method. A simplified request may contain a request line such as **GET / HTTP/1.1** along with headers that provide additional information about the client and the requested resource.

The server processes the request and sends an HTTP response. The response may contain HTML, CSS, JavaScript, images, and other resources. The browser may then send additional requests for these resources before assembling and displaying the complete webpage.

4. HOW TCP PROVIDES RELIABLE COMMUNICATION FOR HTTP

TCP provides a reliable communication service for applications that need accurate and ordered data delivery. Before data is exchanged, TCP establishes a connection using a three-way handshake:

Client → SYN → Server

Server → SYN + ACK → Client

Client → ACK → Server

Connection established

TCP assigns sequence information to transmitted data and uses acknowledgements to confirm successful delivery. It can retransmit missing data, preserve the correct order of received data, regulate the amount of data sent using flow control, and reduce congestion when the network is overloaded. These mechanisms help ensure that webpage content reaches the browser correctly.

5. ARCHITECTURE: CLIENT-SERVER

Architecture Used: Client-Server

The browser acts as the client and initiates requests. Google's web servers receive these requests, process them, and send responses. The server infrastructure can handle requests from many clients at the same time. This is different from a Peer-to-Peer system, where participating devices can directly provide services to one another. Therefore, Google Search follows the Client-Server architecture.

Browser (Client) → HTTP Request → Google Web Server → HTTP Response → Browser displays webpage

6. COMPARISON BETWEEN HTTP AND FTP

HTTP and FTP are both application-layer protocols, but their main purposes and communication styles are different.

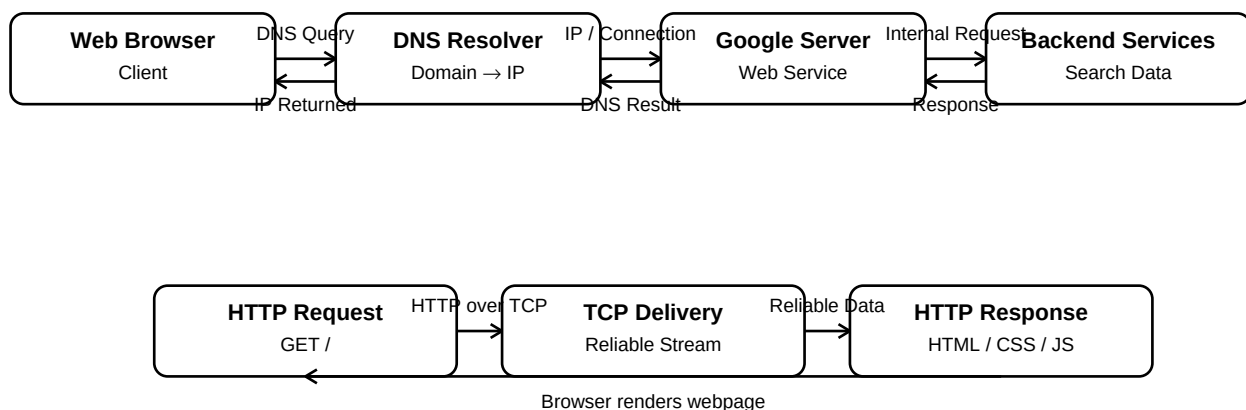
Feature	HTTP	FTP
Main purpose	Used mainly for accessing web resources and communicating with web servers.	Designed primarily for transferring files between a client and a server.
Default ports	Port 80 for HTTP; port 443 for HTTPS.	Port 21 is commonly used for FTP control communication.
Communication style	Uses request–response communication between a client and a web server.	Uses commands and separate file-transfer operations for managing files.
Typical content	Webpages, HTML, CSS, JavaScript, images, and API data.	Files and directories such as documents, images, and software packages.
Common client	Web browsers and web applications.	FTP client software or command-line FTP tools.
Security	HTTPS protects HTTP communication using TLS encryption.	Plain FTP is not encrypted; secure alternatives include FTPS and SFTP.

7. HOW COOKIES IMPROVE USER EXPERIENCE

Cookies are small pieces of information that a website can store in a user's browser. They allow a website to remember selected information between visits and requests. Cookies can maintain login sessions, store preferences, and help a service provide a more convenient experience.

Real-world example: When a user signs in to a Google service, browser-stored information can help the service maintain the user's session and remember selected preferences. As a result, the user may not need to repeat the same login or preference choices on every visit. Cookies therefore make repeated use of web services smoother and more personalized.

8. COMPLETE COMMUNICATION FLOW



URL → DNS → TCP → HTTP Request → Server Processing → HTTP Response → Webpage

The sequence begins with the URL entered by the user. DNS converts the domain name into an IP address. The browser then communicates with the destination server using the TCP/IP protocol stack. HTTP carries the web request and response, while TCP provides reliable transport. The server processes the request and returns the required content, which the browser interprets and renders.

9. CONCLUSION

Application-layer protocols are essential to everyday Internet usage because they provide the services that users directly interact with. When a user opens Google, HTTP or HTTPS enables the browser and web server to exchange requests and responses, while DNS translates a memorable domain name into an IP address. Other application protocols, such as FTP and SMTP, support specialized activities including file transfer and email communication. These protocols depend on the lower layers of the TCP/IP model for reliable transport, logical addressing, routing, and physical delivery. Together, the

layers form an integrated communication system that allows information to travel between a user's device and distant Internet servers. Understanding this process shows how a simple action such as pressing Enter involves multiple protocols working together.

Overall Process: URL entry → DNS resolution → TCP connection → HTTP request → Server processing → HTTP response → Browser rendering